

Snowflake Tutorial Assignment

Full SQL Commands and Steps, Parts 1 to 5

Part 1: SnowSQL Login and Connection

Connect (run this in Command Prompt, not inside SnowSQL):

```
snowsql -a your_account_identifier -u your_username
```

Once connected, run these to verify:

```
SELECT CURRENT_USER();
SELECT CURRENT_ROLE();
SELECT CURRENT_WAREHOUSE();
SELECT CURRENT_DATABASE();
SELECT CURRENT_SCHEMA();
```

Part 2: Creation of Snowflake Objects

```
CREATE DATABASE tutorial_db;
USE DATABASE tutorial_db;
```

```
CREATE SCHEMA tutorial_schema;
USE SCHEMA tutorial_schema;
```

```
CREATE WAREHOUSE tutorial_wh WITH WAREHOUSE_SIZE = XSMALL AUTO_SUSPEND = 60 AUTO_RESUME = TRUE;
USE WAREHOUSE tutorial_wh;
```

```
CREATE TABLE employees (emp_id INT, emp_name STRING, department STRING, salary NUMBER);
```

```
CREATE STAGE tutorial_stage;
```

```
INSERT INTO employees VALUES
  (1, 'Asha Rao', 'Engineering', 55000),
  (2, 'Vikram Nair', 'Sales', 48000),
  (3, 'Priya Menon', 'Marketing', 51000);
```

```
SELECT * FROM employees;
```

```
UPDATE employees SET salary = 60000 WHERE emp_id = 1;
```

```
DELETE FROM employees WHERE emp_id = 3;
```

```
SELECT * FROM employees;
```

Part 3: Data Loading Using SnowSQL

First make a sample CSV file named employees_data.csv using Notepad, with this content:

```
4,Rahul Iyer,Finance,52000
5,Sneha Pillai,Engineering,58000
```

Then run these in SnowSQL (replace YourUsername with your actual Windows username):

```
PUT file:///C:\Users\YourUsername\Documents\employees_data.csv @tutorial_stage;
```

```
COPY INTO employees
  FROM @tutorial_stage/employees_data.csv
  FILE_FORMAT = (TYPE = 'CSV' FIELD_OPTIONALLY_ENCLOSED_BY = ''');
```

```
SELECT * FROM employees;
```

Part 4: Snowflake Time Travel

```
CREATE TABLE time_travel_demo (id INT, value STRING);
```

```
INSERT INTO time_travel_demo VALUES (1, 'Original A'), (2, 'Original B');
```

```
UPDATE time_travel_demo SET value = 'Updated A' WHERE id = 1;
```

```
DELETE FROM time_travel_demo WHERE id = 2;
```

To see the state 5 minutes ago:

```
SELECT * FROM time_travel_demo AT (OFFSET => -60*5);
```

Or using a specific query id from your Query History:

```
SELECT * FROM time_travel_demo BEFORE (STATEMENT => 'your_query_id');
```

Part 5: Data Recovery Using Time Travel

Find the deleted rows by comparing an earlier snapshot to the current table:

```
SELECT * FROM time_travel_demo AT (OFFSET => -60*10)
MINUS
SELECT * FROM time_travel_demo;
```

Recover the missing row:

```
INSERT INTO time_travel_demo
SELECT * FROM time_travel_demo AT (OFFSET => -60*10)
WHERE id = 2;
```

Verify it came back:

```
SELECT * FROM time_travel_demo;
```

Run these in order, and take a screenshot after each part's final SELECT statement showing the output.